

CE 203: Civil Engineering Materials

Course Code	CE 203
Course Name	Civil Engineering Materials
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	B.Tech. (CE)
Mutual Exclusion	None
Approval	50th BoA (Jul 2023), in the B.Tech Civil Engineering curriculum; 52nd BoA (Nov 2023)

Course Contents

- **Fundamentals:** Atomic bonding and molecular structure of materials; Organic, inorganic and metallic construction materials; Alloys and Phase diagrams; Mechanical, physical and thermal properties of materials; Preliminaries of viscoelasticity and rheology; Environmental impact indices for construction materials. (6 Lectures)
- **Ferrous and Non-ferrous metals:** Classification of ferrous alloys; Production process, microstructure and properties of steel; Effects of alloy elements, work hardening and heat treatment on mechanical properties and weldability of steel; Corrosion of steel and its prevention; Structural steel products: Aluminum, Copper and their alloys, production process, properties and uses; Quality and test standards; Sustainability of metals. (6 Lectures)
- **Cement Concrete and Asphalt:** Ingredients of cement concrete and their properties; Concrete mix design and production; Major types of concrete and their characteristics in fresh and hardened states; Durability issues; Cement mortar and its applications; Asphalt binders and their properties; Asphalt mixture formulations; Effects of moisture, temperature and aging on asphalt; Quality and test standards; Sustainability of cement concrete and asphalt. (6 Lectures)
- **Stones, Bricks, Blocks and Tiles:** Composition of good brick earth; Brick manufacturing; Classification of bricks; Classification of rocks; Rock forming minerals; Stone quarrying, seasoning and dressing; Qualities of good building stone and brick; Deterioration and preservation of masonry; Fly ash and concrete masonry units; Sustainability of bricks and blocks; Commonly used tiles for floor, wall and roof; Choosing a floor tile; Quality and test standards. (6 Lectures)
- **Glass Composition of glass:** Production and treatment processes; Physical and mechanical properties; Types and uses; Quality and test standards. (4 Lectures)
- **Wood and Wood products:** Structure of wood, defects and non-uniformities; Physical and mechanical properties; Durability and preservation; Wood-based composites; Quality and test standards; Sustainability of wood. (4 Lectures)
- **Polymers and Plastics:** Classification and properties of polymers; Fabrication methods, additives and fillers used for plastics; Common plastics and their uses; Environmental effects of plastics. (4 Lectures)
- **Paints:** Composition and classification of paints; Production process; Choosing a paint; Defects in painting; Effect of paints on indoor air quality. (3 Lectures)
- **Special topics:** Soil as a construction material; Geosynthetics; Materials for sound and thermal insulation, waterproofing and fire protection; Bamboo; Smart materials; Preliminaries of 3D printing. (3 Lectures)

Text Books

1. Duggal, S.K., **Building Materials**, 4th Edition, New Age International, 2012.
2. Varghese, P.C., **Building Materials**, 2nd Edition, PHI Learning, 2015.

References

1. Young, J.F., Mindess, S., Gray, R., and Bentur, A., **The science and technology of civil engineering materials**, Prentice Hall, 1998.
2. Goncalves, M. C. and Margarido, F., **Materials for Construction and Civil Engineering -- Science: Processing and Design**, Springer Cham, 2015.
3. Subramanian, N., **Building Materials --Testing and Sustainability**, Oxford University Press, India, 2019.